Lattice architecture review

Platform team, first quarter

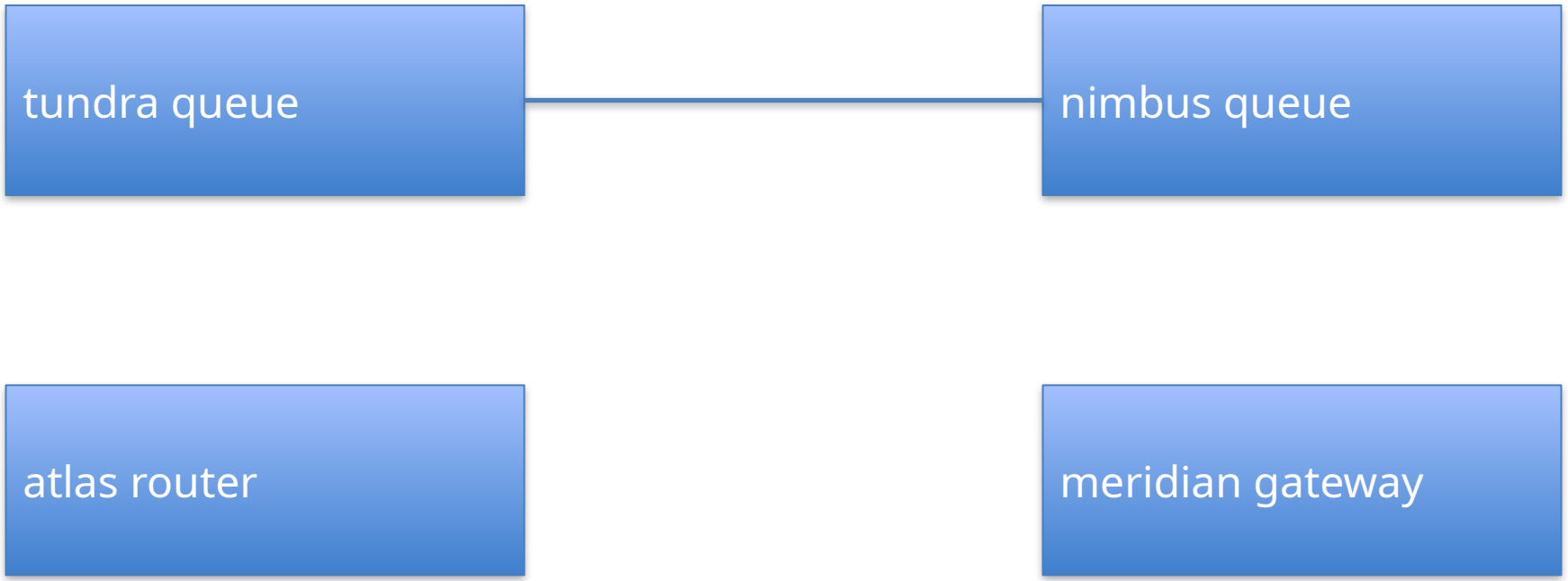
Observability (cobalt)

- Deduplicates pending writes so a failed region drains within minutes
- Rotates index granules after the leader confirms the lease
- Persists incoming events without holding the global lock

Proposed design (harbor)

Option	Latency	Cost	Risk
Option juniper 0	273 ms	$8k / month	medium
Option cobalt 1	224 ms	$37k / month	high
Option ember 2	106 ms	$85k / month	low

Observability (quill)



Timeline (ember)

- Deduplicates malformed payloads when the queue depth passes its threshold
- Signs audit records so that a restart loses no acknowledged work
- Buffers index granules once every replica reports healthy